

SkillsSHEET 4.3

Expanding brackets

To expand brackets, multiply each term in the brackets by the coefficient in front of the brackets.

WORKED Example

Expand each of the following.

a $3(2x - 4)$

b $-2(4x + 7)$

THINK

- a**
- 1 Write the question.
 - 2 Multiply 3 by $2x$ and then 3 by -4 .
 - 3 Simplify.

- b**
- 1 Write the question.
 - 2 Multiply -2 by $4x$ and then -2 by 7 .
 - 3 Simplify.

WRITE

a $3(2x - 4)$
 $= 3 \times 2x + 3 \times -4$
 $= 6x - 12$

b $-2(4x + 7)$
 $= -2 \times 4x + -2 \times 7$
 $= -8x - 14$

Try these

Expand each of the following.

1 $3(2x + 1)$

$= 3 \times \dots + 3 \times \dots$

$= \dots + \dots$

3 $7(x - 5)$

$= \dots \times \dots + \dots \times \dots$

$= \dots - \dots$

5 $4(3x - 1)$

$= \dots$

$= \dots$

7 $-3(3x + 4)$

$= \dots$

$= \dots$

9 $-5(x - 2)$

$= \dots$

$= \dots$

2 $2(4x + 2)$

$= 2 \times \dots + \dots \times \dots$

$= \dots + \dots$

4 $6(5x - 4)$

$= \dots$

$= \dots$

6 $-2(2x + 3)$

$= \dots$

$= \dots$

8 $-4(2x - 2)$

$= \dots$

$= \dots$

10 $-6(x - 1)$

$= \dots$

$= \dots$